InventoryIQ – Project Overview & Architecture Plan

Purpose

InventoryIQ is an enterprise-style inventory optimization platform built to demonstrate cloud analytics, SQL, Python, AWS, and interactive dashboard development. The project simulates a single distribution center using a public logistics warehouse dataset and presents executive-ready analytics.

Objectives

Build a production-style portfolio project; showcase AWS S3, Athena, SQL, Python, Pandas, Plotly, and Streamlit; demonstrate KPI development, inventory optimization, supplier analytics, sales analytics, and risk management.

Architecture

Data Source (Kaggle) → Amazon S3 Data Lake → Amazon Athena (serverless SQL) → Python/Pandas business logic → Streamlit dashboards → Executive insights.

Dashboards

1. Executive Dashboard
2. Inventory Analytics
3. Sales Analytics
4. Supplier Analytics
5. Risk Center

Key KPIs

Inventory Value, Inventory Health Score, Reorder Quantity, High-Risk SKUs, Revenue, Orders, Units Sold, Supplier Reliability, Stockout Risk, ABC Classification.

Project Structure

AWS stores the data, Athena executes SQL queries, Python performs transformations, Streamlit renders interactive dashboards. Shared cached data minimizes reload time.

Future Roadmap

Deploy publicly, add authentication, CI/CD, forecasting, ML demand prediction, automated refresh, executive reporting, monitoring, and custom alerts.

Author

Developed by Sandeep Senthil as a portfolio project targeting Solutions Engineering, Technical Sales, Product Management, Business Analytics, and Cloud Solutions roles.